TEKVOLT SYSTEMS · TRADING ARCHITECTURE

CRYPTALYST

Books, Models & Data

The decoupled streaming forecast architecture —
messaging · streaming · models · books · data · history —
and how they combine to continuously forecast and fire tier signals.

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CryptAlyst — Books, Models & Data

The decoupled streaming forecast architecture

> A complete engineering review of the platform: how **messaging, streaming, models, books, data and history** combine to *continuously forecast* and fire **tier signals** that strategies trade. Every diagram is rendered in the CryptAlyst role palette.

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How to read this document

Each section builds on the previous one, but every section also stands alone.

- **Sections 1–6** establish the pieces: the principle, the bus, the streaming, the models, the books, the data.
- **Sections 7–13** show the flow: forecast → tier signal → trade, the exit ladder, the Go concurrency, the journey, the envelope, the topic registry, the risk governor.
- **Sections 14–25** are the reference: the evolution, each model in detail, worked examples, the data signals, the lineage, the states, the recorder, the supervisor, the risk.
- **Sections 26–34** are the reasoning and the operations: why it is built this way, the forecast math, the code structure, more examples, monitoring, testing, the full registry, and the glossary.

How to read a diagram. Every diagram uses the CryptAlyst role palette:

Colour	Role
blue	market data (raw feed)
green	intelligence (TA, forecast, signals)
amber	models, books, strategies
red	risk, refusals
teal	control, orchestration
purple	stores, the lake
grey	the bus, envelopes, neutral

The colour tells you *what kind of thing* a node is at a glance, before you read a single label. Flow runs left-to-right or top-to-bottom; sequence diagrams run top-to-bottom over time.

Appendix A — the exit ladder in code

The ladder is four checks, in order, on every mark:

```
switch {
case chg >= +0.35:
    reason = "lock"                          // bank a fee-clearing profit
case peak >= +1.0 && chg <= peak-0.5:
    reason = "trail"                         // ride the winner, exit on a pullback
case chg <= -0.5:
    reason = "stop"                          // cap every loss
case held >= 180 && chg > 0:
    reason = "clock"                         // force turnover on winners
}
```

Appendix B — a model rule in code

`rsi2` is the whole model: a pure function of candles, nothing else.

```
func rsi2(bs []Candle) string {
    cl := closes(bs)
    if len(cl) < 3 { return "" }
    r := rsi(cl, 2)
    if r < 10  { return "buy"  }
    if r > 90  { return "sell" }
    return ""
}
```

It holds no state, reads no account, and knows nothing about who consumes its output. That purity is what makes it testable — and what makes it a *calculation agent* rather than a trading loop.

Appendix C — the imbalance in code

```
imb = (bid_volume - ask_volume) / (bid_volume + ask_volume)
if |imb| > 0.3:
    side = buy if imb > 0 else sell
```

One number, from the order book, is the whole curve-rider signal — the BPF "keep only the signal" idea applied to market data.

1. The core principle

What it does. This section defines the one organising idea the whole platform hangs on: every part is a *producer* or a *consumer* of small, stamped packets, and nothing is coupled directly to anything else.

Why it matters. Decoupling is what makes the system observable, testable and replaceable. A coupled system (a model that opens a position directly) is fast to write and impossible to reason about afterwards; a decoupled system records every hop as a packet, so every decision can be replayed and every part can be swapped without touching the rest.

The platform is **asynchronous and decoupled**. Nothing calls anything else directly.

- **Models are calculation agents** — they publish *numbers* (a signal, a score, a probability).
- **The bus carries packets** — every message is an envelope with an **epoch timestamp and a TTL**, and it is **never backlogged**: subscribers read the latest state, nothing queues.
- **Books hold the intel** — a book accumulates the recent signal stream in a **bounded rolling window**, so a model can see a *pattern* before it is confident.
- **Strategies trade** — a strategy lives inside the book and opens/closes only when the model has confirmed a pattern.

A single signal is not enough. A model reports a signal, is *not sure* whether it is a good idea, and waits until it sees a **pattern**. **That intel is what the book carries and the model uses.**

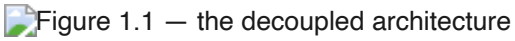


Figure 1.1 — data → models → bus → books → strategy → trade, decoupled at every hop.

1.1 Why decouple

Coupled (what we moved away from)	Decoupled (the design)
a model opens a position directly	a model only publishes a number
one slow model slows everything	each model owns its own goroutine
a panic in one model kills the pipeline	a panic is contained and reported
models only run when the loop runs	models run continuously, whether or not anyone listens
consumers read whatever arrives	consumers read the latest state (TTL, no backlog)

The decoupling is what makes the system *agile*: a packet of model output or market data can be published, listened to, and used by any other part — without any part knowing who listens.

2. Messaging — the bus is the nervous system

What it does. The bus moves every packet between every component: market data in, model numbers out, decisions and fills and pnl around the loop. A packet is an envelope — a payload plus provenance (who, when, in what order, with a TTL).

Why it matters. The bus is the single point of integration. Because everything flows through it as stamped envelopes, the whole system has one audit trail, one source of truth for "who may write what", and one place to observe. Without it, every component would need its own hand-rolled wiring, and the platform would be a hairball of direct calls.

Every message on the bus is an **envelope**: a payload plus metadata — the provenance that makes the stream auditable and fresh. The bus is a **state channel, not a job queue**: with a TTL on every packet and no backlog, a subscriber always reads the *current* picture, and old packets simply evaporate.

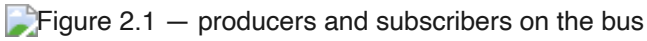


Figure 2.1 — calculation agents publish numbers; subscribers listen and decide.

2.1 The envelope anatomy

```
flowchart TD
    E["envelope"] --> D["data — the payload"]
    E --> M["meta — version · plane · producer · epoch ts · TTL · cid"]
    E --> R["refs — topic · key · generation · age_ms"]
```



Figure 2.2 — an envelope carries the payload, its provenance, and the inputs it read.

Member	What it carries	Why it matters
`data`	the payload for the topic	the number the model computed
`meta`	version, plane, producer, generation, cid, timestamps	provenance — who, when, in what order
`refs`	the inputs it read (topic, key, generation, age)	lineage — what a decision was based on

2.2 The metadata fields

Field	Meaning
`version`	the envelope contract revision
`plane`	market, intel, model, risk, control, system
`topic`	the channel (e.g. `model.decision`)
`key`	the symbol, when the message concerns one
`producer`	the component that computed the payload
`generation` / `seq`	monotonic ordering per topic
`cid` / `parent_cid`	correlation id — links a decision to its fills
`computed_at`	when the payload was produced (staleness gates use this)
`received_at`	when the recorder accepted it
`hops` / `refs`	the pipeline stages and the inputs read

2.3 The pipeline — interceptors, operator, transformer



Figure 2.3 — authority, keying, correlation and freshness run before the operator produces a payload.

Every envelope passes four interceptors before the operator runs, and a transformer after:

Stage	What it checks	Refusal records
`authority`	the producer may write this topic	the rule name
`keying`	the message is keyed by symbol	the rule name
`correlation`	a cid is minted and chained	the rule name
`freshness`	`computed_at` is within the topic's limit	the rule and the number

A refusal is recorded as an envelope too — so "why not" is a packet you can open, not a line in a log.

2.4 Topics — a single registry with producer contracts

`topics.go` is the **one table**. Each topic declares its exchange, its producers and its retention (TTL). The planes group them:

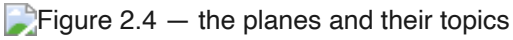


Figure 2.4 — five planes: market, intel, model, risk, control.

Topic	Exchange	Producers	Retention
`market.price` / `market.depth` / `market.candle`	market	market_data	drop (live only)
`intel.signal` / `intel.flow` / `intel.movers`	intel	signals	60s
`intel.ta`	intel	tad	60s
`intel.forecast`	intel	forecast	60s
`model.decision`	model	models10, curve_rider, …	5m
`model.fill`	model	same	5m
`model.pnl`	model	same + supervisor	5m
`risk.state`	risk	risk	5m

3. Streaming — never poll (BPF-inspired)

What it does. Market data (depth, candles, price, volume, OI) arrives over WebSocket, pushed by the exchange, and is filtered at the source to only the fields each consumer needs.

Why it matters. Polling is the enemy of a fast decision: the measurement showed one HTTP fetch (333 ms) is ~2000× slower than one model calculation (150 μs). Streaming removes that latency and the "filter at the source" idea (from the Berkeley Packet Filter) removes the wasted data. A decision made on a pushed, pre-filtered number is both faster and cheaper.

The Berkeley Packet Filter filters packets **at the source** — inside the kernel — so only the data that matches is ever copied up. The same idea applied to market data:

1. **Filter at the source** — subscribe to only what we need (`depth5`, not the full book).
2. **Keep only the signal** — collapse the order book to one `imbalance` number.
3. **Never poll** — data pushes to us over WebSocket; decisions happen on arrival.

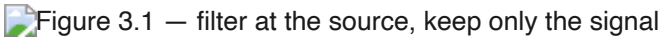


Figure 3.1 — WebSocket → filter → one number, the BPF pattern.

3.1 Why streaming beat polling

Measured during the build: **one model calculation takes ~150 µs**, while **one HTTP fetch takes ~333 ms** — the network was ~2000× slower than the math. Polling 24 endpoints sequentially cost ~8 seconds per decision; the WebSocket stream delivers the same data in real time.

Before (poll)	After (stream)
24 sequential HTTP fetches ≈ 8s per decision	push, ~0 latency
full depth (20 levels)	depth5 (5 levels) + one imbalance number
stale data on a timer	data on arrival

The bottleneck was never the language — it was I/O and polling.

3.2 What is streamed

Stream	Frequency	What it feeds
`depth5@100ms`	100 ms	the imbalance (book pressure)
`kline_1m` / `kline_15m`	on close	the TA models
`bookTicker`	real time	the best bid/ask
`trade` / `aggTrade`	real time	volume and taker flow

The market data service subscribes once, and every model reads the same live state.

4. Models — calculation agents that publish numbers

What it does. A model is a *pure function*: candles and the depth imbalance in, a "buy" / "sell" / "nothing" out. It publishes that number and holds no account state.

Why it matters. Purity is what makes a model testable (you can feed it generated bars and assert the answer) and replaceable (a better rule is a drop-in). If a model also knew about the account, it would be impossible to test in isolation and impossible to swap without dragging the whole book along. One model = one question = one testable function.

A model owns **one question**. It does not know about accounts, positions, or risk. It computes and publishes — which is what makes it replaceable and testable.

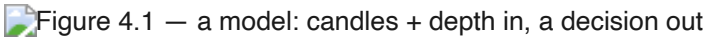


Figure 4.1 — the model consumes the cached market state, runs its rule, and publishes a decision.

4.1 The models

Model	Question	Rule
`donchian55`	breaking a 55-bar range?	close ≥ 55-bar high → buy; ≤ low → sell
`orb`	breaking the opening range?	close > session high → buy; < low → sell
`boll_squeeze`	tight Bollinger expanding?	close > +2σ with width < 1% → buy
`rsi2`	2-bar RSI stretched?	RSI < 10 → buy; > 90 → sell
`vwap_revert`	stretched from VWAP?	below 0.994×VWAP → buy; above 1.006× → sell
`macd_trend`	MACD crossed zero?	hist crosses up → buy; down → sell
`vol_break`	volume spike + directional close?	vol > 3×avg + close in top/bottom 30% → trade
`ema_pullback`	pullback to the EMA trend?	price at EMA20 with EMA20>EMA50 → buy
`atr_channel`	outside an ATR channel?	close > EMA20+1.5×ATR → buy
`curve_rider`	strongest book imbalance?	buy/sell the biggest bid/ask imbalance



Figure 4.2 — RSI2, Donchian and MACD: each is a tiny deterministic function of candles.

4.2 Models feed other models

A model's output is input to another. A **positioning** model publishes where the book is exposed; a **risk** model consumes it and publishes risk; both feed the decision. **Models provide valuable data for other models** — and the bus is how that data moves without any model knowing who listens.

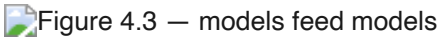


Figure 4.3 — positioning → risk → decision, each a published number.

5. Books — account and intel

What it does. The book is the account (equity, positions, PnL) *plus* a bounded rolling window of the recent signals, so a model can see a *pattern* before it is confident.

Why it matters. A single signal is noise; a streak is a signal. The window is the memory that turns the former into the latter — without it the strategy would trade every blip. Because the window is bounded and TTL-expired, it never holds too much, yet it is long enough to catch a sustained move (imagine a bull run: the streak is exactly what you want to see).

The book is two things at once:

1. **The account** — equity, open positions, realised PnL. This survives and is reported.
2. **The intel** — a bounded rolling window of recent signals, so the model can see a *pattern* rather than a lone blip.

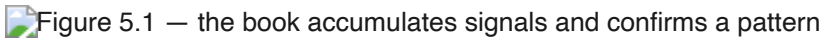


Figure 5.1 — a lone signal is noise; the streak in the window is the signal.

5.1 The pattern confirmation

Models need **previous signals, or many**, to make an informed decision. The book holds those for a **time window the model defines**:

- a signal arrives — the model is *not sure* yet;
- the book records it in the rolling window;
- only when **N agreeing signals** accumulate inside the window does the model confirm;
- the confirmation becomes a **tier signal** the strategy may act on.

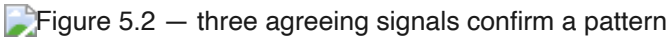


Figure 5.2 — a signal is reported; the model waits for a pattern before it is sure.

"Never hold too much data, but imagine a bull run." The window is bounded — old entries expire — yet it is long enough that during a sustained move the model sees the *streak* of agreeing signals that confirms the trend. A single signal is noise; the **streak** is the signal.

6. Data — the market microstructure

What it does. The system reads the order flow — depth imbalance, open interest, price, volume — not just the candles. Each signal reveals a different face of what the market is *thinking*.

Why it matters. Candles alone are the *past*; the order book and OI are the *present* — the resting orders and the new money that will move price next. Combining them with history turns a lagging picture into a leading forecast. The market microstructure is the edge: it shows intention before it becomes a completed trade.

A forecast is only as good as the data behind it. The system reads the order flow, not just the candles:

Signal	Reads	What it reveals
depth imbalance	bid vs ask volume	real buy/sell pressure now
OI change	open interest over an interval	new money entering
price	last / 24h move	direction + momentum
volume	quote volume / spike	participation and conviction

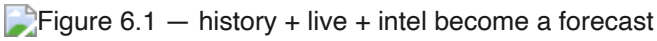


Figure 6.1 — three layers combined turn a number into a forecast.

6.1 The three layers of data

Layer	What it provides	Why it matters
Live stream	the *now* — imbalance, last price, last volume	direction of pressure
Historical bars	the *context* — 200 bars of OHLCV	where price is in its range
Signal intel	the *memory* — recent model outputs	whether a pattern is forming

All three combined turn a number into a *forecast*, and a forecast into a **tier signal**.

7. Forecast → tier signal → trade

What it does. The numbers from the models are combined with history and the signal intel into a *forecast*, which is then ranked into *tiers*, and only a confirmed tier signal is traded.

Why it matters. Not every number deserves a trade. The tier ranking is what stops the strategy from treating a weak signal and a strong one the same — it concentrates capital on the confident, confirmed signals and ignores the noise. That is the difference between a forecast and a firehose.

The strategy does not act on every number. It waits for a **tier signal** — a confirmed, strength-ranked decision.

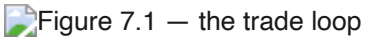


Figure 7.1 — numbers → pattern → tier signal → open → manage → fill.

7.1 Tier ranking

Not all signals are equal. The system ranks them by strength and acts accordingly:

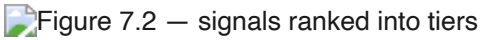


Figure 7.2 — tier 1 acts immediately, tier 2 needs confirmation, tier 3 is ignored.

Tier	Meaning	Action
Tier 1	strong, confirmed	act immediately
Tier 2	medium	needs further confirmation
Tier 3	weak	ignore / log only

8. The strategy and exit ladder

What it does. Once a tier signal is confirmed, the strategy opens a position and manages it on a fixed ladder: lock at +0.35%, trail off the peak, stop at –0.5%, clock at 180s.

Why it matters. The ladder is a *risk envelope*, not a prediction. It does not try to call the market; it guarantees that whatever happens, the profit is banked, the loss is bounded, the winner has room to run, and nothing is held forever. Fixed, simple exits are what make the outcome predictable — which is the whole point of a trading system.

Once a tier signal is confirmed, the strategy opens a position and manages it on a fixed ladder. The ladder is the risk envelope that keeps losses bounded and lets winners run.

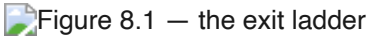


Figure 8.1 — four exits, checked in order, on every mark.

Exit	Rule	Purpose
`lock`	+0.35%	bank a fee-clearing profit
`trail`	peak ≥ +1%, exit –0.5% off peak	ride winners
`stop`	–0.5%	cap every loss (no heavy bleeds)
`clock`	180s hold, winners only	force turnover

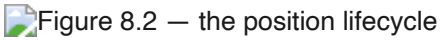


Figure 8.2 — open → mark → exit ladder → close → fill.

9. Pure Go — concurrency

What it does. The models and the engine run as independent Go goroutines inside one process, each with its own scheduler.

Why it matters. The same independence the bus gives at the message level, goroutines give at the process level: a slow model can't block the others, a panic is contained, and each model runs continuously whether or not anyone listens. Pure Go also removes the interpreter as a concern — the math was never the bottleneck, but one runtime for everything is simpler to build, deploy and reason about.

The models and the engine run as **Go goroutines**, each decoupled from the others:

- **A slow model can't block the others** (each owns its goroutine).
- **A panicking model can't kill the pipeline** (panic is contained and reported).
- **Models run continuously**, publishing numbers whether or not anyone listens.

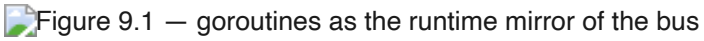


Figure 9.1 — the same independence the bus gives at the message level, goroutines give at the process level.

10. The full journey

What it does. Traces one market tick from the exchange WebSocket all the way to a recorded fill, through the bus, the recorder, the models, the book and the lake.

Why it matters. The journey is the proof the decoupling works end to end: data is recorded once and consumed many times, models publish and books decide, and every step lands in the same append-only lake. Understanding the journey is understanding the whole system in one pass.

One market tick's journey through the platform, end to end:



Figure 10.1 — data is recorded once and consumed many times; models publish, books decide.

Stage	What happens
1 · market data	Binance WebSocket → market.price / market.depth / market.candle
2 · the bus	every packet is an envelope with epoch ts and TTL
3 · recorder	the single consumer; writes the lake once
4 · models	subscribe, compute, publish model.decision
5 · book	accumulates the intel, confirms a pattern
6 · strategy	opens/closes, publishes model.fill
7 · lake	the fill is recorded, joining the same history

11. The envelope and pipeline

What it does. Defines the envelope (data, meta, refs) and the pipeline every packet passes: four interceptors, the operator, a transformer.

Why it matters. Provenance and checks are *on the packet*, not in a log beside it. A refusal is recorded as an envelope too, so "why not" is a packet you can open. That is what turns a message stream into evidence you can audit.

Covered in section 2 — reproduced here with the full field dictionary and the worked refusal path.

11.1 Reading a decision

A `model.decision` envelope carries the verdict, the confidence, and the reason. A refusal names the rule and the number that tripped it. The pipeline shows the interceptors that ran.

11.2 The refusal path

A refusal is not a log line — it is an envelope you can open:

```
model.decision NEARUSDT
  verdict   REFUSED
  reason    stale signals: 3230s old (max 90s)
  pipeline  authority pass · keying pass · correlation pass · freshness pass
  operator  alpha_rank · produced model.decision
  refs      market.candle 5007ms (derived) · market.depth 906ms (derived)
```

The freshness interceptor passed because the *envelope* was fresh; the refusal came from the model's own signal-age check — a distinction the record preserves.

12. The topic registry

What it does. One table lists every topic: its exchange, its allowed producers, and its retention (TTL).

Why it matters. A single registry is the enforcement point for producer contracts — who may write what — and for the TTL that keeps the bus a state channel rather than a job queue. Adding a topic is one line; changing a contract is visible to everyone.

The registry is the single source of truth for who may write what, and how long it lives.

Topic	Plane	Producers	Retention
`market.price`	market	market_data	drop
`market.depth`	market	market_data	drop
`market.candle`	market	market_data	drop
`market.mini24h`	market	market_data	drop
`ticks`	market	market_data	drop
`intel.signal`	intel	signals	60s
`intel.flow`	intel	signals	60s
`intel.movers`	intel	market_data, signals	60s
`intel.board`	intel	market_agent	60s
`intel.ta`	intel	tad	60s
`intel.forecast`	intel	forecast	60s
`alert.pumpdump`	alert	backend, signal_bus	300s
`model.decision`	model	models10, curve_rider, …	5m
`model.fill`	model	same	5m
`model.pnl`	model	same + supervisor	5m
`risk.state`	risk	risk	5m

Why TTL + no backlog: a market moves on the latest information; the bus is a state channel, not a job queue.

13. The risk governor

What it does. Sits between the tier signal and the trade, and refuses any order the account cannot afford: exposure, gross multiple, per-symbol concentration.

Why it matters. The models want to trade; the account has limits. The governor is the machine-enforced boundary that says no, with the rule and the number recorded. Without it, "risk" would be a hope; with it, a refusal is a fact you can audit.

Between the tier signal and the trade sits the risk governor. It is the machine-enforced boundary that refuses a trade the models want but the account cannot afford.



Figure 13.1 — every decision passes the exposure and gross-multiple checks before an order.

Check	What it refuses
exposure	a position above a fraction of the book
gross multiple	total notional beyond a leverage bound
per-symbol	concentration in one instrument
counter-trend	shorts into an up-regime (when disabled)

A refusal names the rule and the number, and is recorded — so "why not" is always answerable.

14. The evolution

What it does. Recounts the path from a monolithic script to the decoupled streaming forecast.

Why it matters. The evolution is the design argument in miniature: each step removed a coupling and added a layer of memory. Seeing the steps explains *why* the end state looks the way it does — and warns against sliding back into the coupled pattern that caused the trouble in the first place.

Every step removed a coupling and added a layer of **memory** — which is exactly what a forecast needs.

Step	What changed
Monolithic script	models opened positions directly (coupled)
Streaming	depth + candles over WebSocket (never poll)
Pure Go	TA rules + book ported to Go (goroutines)
Decoupled bus	models publish model.decision over STOMP
Intel + patterns	book holds a signal window, confirms before trading
Continuous forecast	history + live + intel → tier signals

15. Model reference — the calculation agents in detail

What it does. Documents each of the ten models as a pure function — its exact rule, its inputs, its thresholds, and what it returns.

Why it matters. The reference is the contract between "what the model computes" and "what the system trades". Having each rule written down means a signal can be reproduced by hand, a bug can be localised to one function, and a change can be reasoned about before it is made.

Each model is a **pure function of its inputs**: candles (and, for the curve rider, the depth imbalance). It returns `buy`, `sell`, or nothing. It holds no state and knows nothing about accounts. That purity is what makes every model testable against generated data.

15.1 `donchian55` — range breakout

Question: is price breaking out of a 55-bar range?

```
hi = max(high[-55:]);  lo = min(low[-55:])
if hi == lo: return none
c = close[-1]
if c >= hi: return buy
if c <= lo: return sell
return none
```

Input	Lookback
55-bar high/low	55 candles
close	last candle

A breakout is the oldest and most persistent edge: a market that leaves its recent range tends to keep going. The `hi == lo` guard rejects a flat symbol, where no breakout is possible.

15.2 `orb` — opening range breakout

Question: is price breaking the first 15 minutes of the UTC day?

```
session = candles from today
if len(session) < 20: return none
hi = max(high of first 15);  lo = min(low of first 15)
if close > hi: return buy
if close < lo: return sell
```

Input	Lookback
first 15 candles of the day	session start
close	last candle

The opening range is where the day's conviction is established; breaking it is a genuine directional commitment rather than intraday noise.

15.3 `boll_squeeze` — volatility expansion

Question: is a tight Bollinger band starting to expand?

```
cl = close[-20:];  m = mean(cl);  sd = stdev(cl)
width = 2*sd/m
if close > m + 2*sd and width < 0.01: return buy
if close < m - 2*sd and width < 0.01: return sell
```

Input	Threshold
20-bar mean + stdev	width < 1% (squeeze)
close outside ±2σ	breakout

A squeeze (narrow band) precedes a move; the breakout direction is the signal. The width guard ensures the model only fires when the market was genuinely compressed.

15.4 `rsi2` — mean reversion at the extreme

Question: is the 2-bar RSI overbought or oversold?

```
r = rsi(close[-20:], period=2)
if r < 10: return buy
if r > 90: return sell
```

Input	Threshold
2-bar RSI	< 10 → buy; > 90 → sell

A very short RSI measures momentum exhaustion. The 2-bar period makes it respond in seconds, so it catches the exact moment a move is spent.

15.5 `vwap_revert` — mean reversion to volume

Question: is price stretched from its volume-weighted average?

```
vw = sum(close*vol over 30 bars) / sum(vol over 30 bars)
if close < vw * 0.994: return buy
if close > vw * 1.006: return sell
```

Input	Threshold
30-bar VWAP	±0.6% band

Price stretched 0.6% from VWAP is pulled back; the model fades the stretch.

15.6 `macd_trend` — momentum cross

Question: did the MACD histogram cross zero?

```
fast = ema(close, 12);  slow = ema(close, 26)
hist = fast - slow
prev = ema(close[:-1], 12) - ema(close[:-1], 26)
if prev <= 0 < hist: return buy
if prev >= 0 > hist: return sell
```

Input	Threshold
MACD(12,26) histogram	cross above/below zero

The zero cross is the moment short-term momentum overtakes the trend — a clean, binary turn signal.

15.7 `vol_break` — volume confirmation

Question: is volume spiking with a directional close?

```
avg = mean(volume[-20:-1])
if volume[-1] < 3*avg: return none
rng = high[-1] - low[-1]
if (close-low)/rng > 0.7: return buy
if (high-close)/rng > 0.7: return sell
```

Input	Threshold
volume vs 19-bar average	> 3×
close in the bar's range	top/bottom 30%

A volume spike that closes near the extreme of the bar is participation with conviction — the difference between a move and a wiggle.

15.8 `ema_pullback` — trend continuation

Question: is price pulling back to the EMA trend and holding?

```
e20 = ema(close, 20);  e50 = ema(close, 50)
if e20 > e50 and close > e50 and |close−e20|/e20 < 0.002: return buy
if e20 < e50 and close < e50 and |close−e20|/e20 < 0.002: return sell
```

Input	Threshold
EMA20 vs EMA50	trend
close vs EMA20	within 0.2%

A pullback to a rising EMA that holds is the classic continuation entry — with the trend, at a discount.

15.9 `atr_channel` — volatility channel breakout

Question: is price outside an ATR-based channel?

```
m = ema(close, 20);  a = atr(14)
if close > m + 1.5*a: return buy
if close < m − 1.5*a: return sell
```

Input	Threshold
EMA20 ± 1.5×ATR	channel

The channel adapts to volatility: in a quiet market the channel is tight, in a wild one it is wide, so "outside the channel" always means a real move.

15.10 `curve_rider` — order-flow imbalance

Question: which market has the strongest book imbalance, and which way?

```
imb = (bid_volume − ask_volume) / (bid_volume + ask_volume)
if |imb| > 0.3:
    side = buy if imb > 0 else sell
```

Input	Threshold
depth5 bid/ask volume	limbl > 0.3
price	best bid/ask midpoint

The order book is what the market is *thinking* right now — heavier bids mean real buying pressure, heavier asks real selling. The curve rider enters the strongest and rides it with the trail.

16. Worked examples

16.1 A bull run — how the streak becomes a tier signal

During a sustained up-move, a single `donchian55` buy signal is not enough. The book records each signal in its rolling window:

Time	Signal	Window state	Decision
t0	buy	[buy]	wait — 1 signal
t1	buy	[buy, buy]	wait — 2 signals
t2	buy	[buy, buy, buy]	**pattern confirmed → tier signal**

The model was *not sure* at t0 and t1; at t2 the streak made it confident. The strategy opens, and the `trail` exit lets the position run while the streak holds — the bull run is ridden, not skimmed at the first tick.

16.2 A refusal — why the model did nothing

A model presents a decision, but the risk governor refuses it:

```
model.decision SOXLUSDT  side=buy  confidence=0.71
  governor   REFUSED  gross multiple 12.4× > limit 8×
```

The refusal names the rule and the number. Nothing is silent: the refusal is recorded as an envelope, and the WHY view lists it with the exact reason. The system did not fail to trade — it correctly refused a trade the account could not afford.

16.3 A full trade — entry to fill

```
t0  pattern confirmed (3 × donchian55 buy)          → strategy opens 2500 notional
t1  mark +0.41%  → lock (≥ +0.35%)                  → close, net +0.21%
    → publish model.fill {reason: lock, net_pct: 0.21}
t2  the fill joins the lake, correlated to the decision
```

The whole chain — decision → fill → pnl — is linked by one correlation id, so any trade can be traced from the market tick that triggered it to the fill it produced.

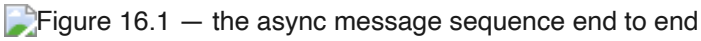


Figure 16.1 — marketdata → bus → model → book → strategy → recorder, one tick's journey.

17. Data signals — deep dive

What it does. Explains the three microstructure signals in detail: depth imbalance, open interest change, and volume — each with its formula and its thresholds.

Why it matters. These signals are what the market is *thinking*, as opposed to what it has *done* (the candles). Understanding each one — and its limits — is what lets the models read intention rather than lag. The depth leads price; OI shows conviction; volume shows participation.

17.1 Depth imbalance — what the book is thinking

The order book is a live vote. Bid volume is money wanting to buy; ask volume is money wanting to sell. The imbalance is the net of the two, normalised:

```
imbalance = (bid_volume − ask_volume) / (bid_volume + ask_volume)
```

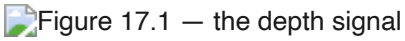


Figure 17.1 — heavier bids → long; heavier asks → short; balanced → no signal.

imbalance	Reading	Action
> +0.3	buy pressure	consider long
< –0.3	sell pressure	consider short
near 0	balanced	no signal

The book is what the market is *thinking right now* — it leads price, because it shows the orders already resting that will move the market next.

17.2 Open interest — new money entering

Open interest is the number of open contracts. A change in OI over an interval tells you whether a move is backed by *new* positions (conviction) or just unwinding (exhaustion).

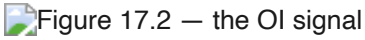


Figure 17.2 — OI rising with price = new longs; OI rising against price = new shorts.

OI	Price	Reading
rising	rising	new longs — continuation up
rising	falling	new shorts — continuation down
flat	either	no new conviction

17.3 Volume — participation and conviction

Volume is participation. A price move on high volume is conviction; the same move on thin volume is noise. The spike rule catches the difference:

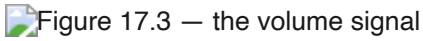


Figure 17.3 — a spike with a directional close is conviction; a spike with a middle close is churn.

Volume	Close position in bar	Reading
> 3× average	top 30%	buy
> 3× average	bottom 30%	sell
> 3× average	middle	churn — no conviction
normal	any	no signal

18. Correlation and lineage

What it does. Explains how one correlation id follows a piece of work from the market tick through the decision, the fill and the pnl, and how references are declared or derived.

Why it matters. Lineage is what turns a pile of packets into a *story* — and a story you can verify. A trade traceable to its trigger is evidence; a trade without a line of descent is a rumour. The correlation id is the spine of the audit.

Every piece of work is followed by one **correlation id**. A market tick leads to a decision, the decision to a fill, the fill to a pnl update — all carrying the same cid, so the whole chain is resolvable in the lake.

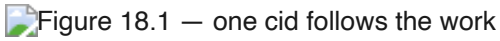


Figure 18.1 — tick → decision → fill → pnl, all keyed to correlation id A.

The lineage is what turns the platform into evidence: any trade can be traced from the raw market packet that triggered it to the fill it produced, with no interpretation required.

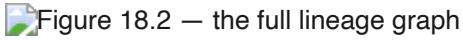


Figure 18.2 — market data → decision → (fill | refusal) → pnl.

18.1 Declared vs derived references

A reference is **declared** when the producer names its inputs; it is **derived** when the recorder infers them (the newest envelope of each input topic for that symbol in the window, marked with its age). The record always says which — a derived reference is never silently presented as declared.

19. Book states

What it does. Models the book as a small state machine: idle → accumulating → confirmed → in position → closed → idle.

Why it matters. A state machine makes the book's behaviour finite and testable. The window expiring returns "accumulating" to "idle", so a pattern can never form from stale intel. Every transition is a defined, checkable step — which is what makes the book deterministic.

The book is a small state machine. Every position moves through the same states, and the exit ladder is the only thing that moves it forward.

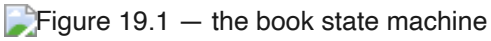


Figure 19.1 — idle → accumulating → confirmed → in position → closed → idle.

State	Meaning
`Idle`	no position, waiting for a signal
`Accumulating`	signals arriving, pattern not yet confirmed
`Confirmed`	N agreeing signals in the window
`InPosition`	a position is open, being marked
`Closed`	the position closed, fill recorded

The window expiring returns `Accumulating` to `Idle` — old signals are dropped, so a pattern never forms from stale intel.

20. The recorder and the lake

What it does. The recorder is the single consumer of the bus: it writes every envelope once into the lake (the append-only journal + an indexed store) and fans it out to readers.

Why it matters. Single-writer, append-only is what makes the record trustworthy: nothing rewrites history, nothing drops a packet silently, and the explorer reads the same lake the recorder wrote. The lake is the memory of the whole platform.

The bus has **one consumer**: the recorder. It writes every envelope **once** into the lake, then fans it out to readers (the explorer, the dashboards). This single-writer design is why the record is append-only and auditable.



Figure 20.1 — bus → recorder → journal + indexed store → explorer.

Store	What it holds
`envelopes.jsonl`	the append-only journal
sqlite / mongo	the indexed, queryable copy
hub	fans envelopes to readers

21. The supervisor and control plane

What it does. The supervisor launches and restarts the models, watches their health, publishes the aggregate pnl, and accepts control commands.

Why it matters. The supervisor is the orchestrator that never trades: it keeps the trading processes alive and honest without making a single decision itself. Separating control from trading means a control failure can't corrupt a book, and a book failure can't silence the control plane.

The supervisor is the orchestrator: it launches and restarts the models, watches health, and publishes the aggregate pnl. It never trades — it only keeps the trading processes alive and honest.

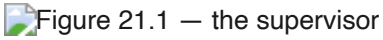


Figure 21.1 — launch/restart models, watch health, publish pnl, accept control.

Duty	Topic / action
launch + restart models	process supervision
watch health	`system.health`
publish aggregate pnl	`model.pnl`
accept commands	`system.control`

22. Example trades — each exit, worked

What it does. Works each exit in the ladder — stop, trail, cascade, funding — as a concrete trade with numbers.

Why it matters. Each exit exists for a reason (bound the loss, ride the winner, buy the flush, collect the funding), and seeing the numbers makes the reason concrete. The ladder is a set of *guarantees*, and each worked example is one guarantee demonstrated.

22.1 The stop

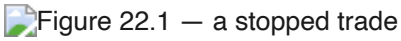


Figure 22.1 — a –0.55% mark trips the –0.5% stop, capping the loss.

The stop is the non-negotiable floor. On 2,500 notional, a –0.5% gross move is –0.7% net after fees — a loss of about $18. The point of the stop is not that it is small, but that it is *bounded*: no position is allowed to bleed beyond it.

22.2 The trail

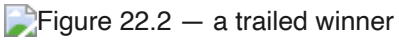


Figure 22.2 — the position peaks +1.4%, arms the trail, and exits on a −0.5% pullback at +0.9%.

The trail is how a winner is ridden: once the peak clears +1%, the exit follows the price down, so the position stays in while the move holds and exits only when it genuinely turns.

22.3 The cascade rebound

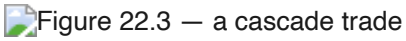


Figure 22.3 — a –3σ drop on 5× volume is bought, and the snap-back is ridden.

The cascade model buys the liquidation flush — the sharpest, most frightened sellers — and rides the reflexive bounce.

22.4 The funding carry

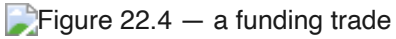


Figure 22.4 — negative funding means shorts pay longs; the book longs and collects.

When funding is deeply negative, shorts are paying longs to hold. The book goes long those perps, collects the funding, and exits on the clock or the lock.

23. Risk accounting

What it does. Measures the account as a whole — the gross multiple (total notional over equity), per-position exposure, and per-symbol concentration — and refuses past the limits.

Why it matters. Risk is a portfolio property, not a per-trade property. Ten individually small positions can still be a lethal total; the gross multiple catches that. The refusal names the rule and the number, so the bound is auditable, not aspirational.

The governor measures the account as a whole, not trade by trade. The binding number is the **gross multiple**: total notional divided by equity.

```
gross multiple = Σ notional / equity
```

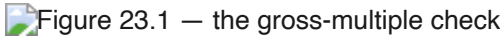


Figure 23.1 — total exposure is measured against equity, and refused past the limit.

Check	Refuses when
gross multiple	total notional exceeds the leverage bound
exposure	one position is too large a fraction of the book
per-symbol	one instrument is over-concentrated

The refusal names the rule and the number — `gross 12.4× > 8×` — so the bound is never a vague feeling but a recorded, auditable decision.

24. The envelope journey, restated

What it does. Restates the pipeline every envelope passes: interceptors → operator → transformer → lake, with a refusal recorded at any stage.

Why it matters. Repetition is deliberate: the pipeline is the one thing every packet shares, so it bears repeating. A refusal at any interceptor is itself an envelope, so the record contains both the trades and the reasons for non-trades.



Figure 24.1 — authority, keying, correlation, freshness, then the operator, then the transformer.

A refusal at any interceptor is itself recorded as an envelope, so the record contains both the trades *and* the reasons for non-trades.

25. Tier decisions, restated

What it does. Restates how numbers are ranked into tiers, and how the tier decides the action.

Why it matters. The tier ranking is the capital-allocation rule: strong confirmed signals act, weak ones are ignored. Restating it alongside the forecast closes the loop — the forecast produces the number, the tier decides whether it is worth trading.

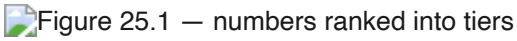


Figure 25.1 — strength + confirmation decide the tier, and the tier decides the action.

The tier system is what prevents the strategy from treating every number as equal: a strong confirmed signal acts immediately, a weak one is ignored. That ranking is the difference between a forecast and a firehose.

26. Design decisions — why it is built this way

What it does. States each major design choice and its reason: the bus over direct calls, the intel window, the TTL, the fixed ladder, the pure models.

Why it matters. Decisions should be written down, or they get re-litigated (and accidentally reversed) later. This section is the "why" appendix — the reasoning that survives the code.

Every choice is a decision with a consequence. These are the reasons, stated plainly.

26.1 Why the bus and not direct calls

Decision	Reason	Consequence
models publish numbers	a model must not know who listens	any consumer can use any model's output
envelopes with TTL	the market moves on the latest state	stale packets evaporate; no backlog
a single topic registry	one source of truth for who may write what	a producer contract is one line
correlation ids	follow the work across components	any trade is traceable end to end

The alternative — a model calling a book directly — is faster to write and impossible to reason about. The bus trades a little latency for complete observability.

26.2 Why the book holds the intel

A model that acts on one signal is a coin flip. The same model that waits for a *streak* is a forecast. The book's rolling window is the memory that turns the former into the latter.

Without the window	With the window
one signal → trade	N agreeing signals → trade
noise is traded	noise is filtered
the model is a blip detector	the model is a pattern detector

26.3 Why TTL and no backlog

A queued message is a stale message. If a slow subscriber read a queue, it would act on data that is minutes old. With TTL, the bus is a **state channel**: a subscriber reads the current picture or nothing. The freshness interceptor enforces the same idea per-envelope.

26.4 Why the exit ladder is fixed

A fixed ladder is a *risk envelope*, not a strategy. It says: profit is banked at +0.35%, loss is capped at –0.5%, winners are ridden, and nothing is held forever. The ladder does not predict the market — it ensures that whatever the market does, the outcome is bounded and the winners have room to run.

26.5 Why the models are pure

A pure function of candles + depth is testable against generated data — no network, no clock, no state. That is what the model test harness relies on: every rule is a function, so every rule can be verified with synthetic bars before it ever sees a live market.

27. The forecast — how the layers combine

What it does. Defines the forecast as a function of three layers — live (the trigger), historical (the context), and signal intel (the confidence) — and shows a worked combination.

Why it matters. A forecast is not one number; it is the combination of *now*, *where we are*, and *is this real*. Separating the layers makes the forecast explainable — you can say which layer drove a decision, and tune each layer independently.

The forecast is not one number. It is a function of three layers:

```
forecast = f( live · historical · signal intel )
```

Layer	Contribution	Weight in the decision
live	direction of pressure now	the *trigger*
historical	where price is in its range	the *context*
signal intel	whether a pattern is forming	the *confidence*

27.1 The trigger — live

The live layer answers "what is happening now": the depth imbalance, the last price, the volume spike. It is the fastest and the noisiest, which is why it alone is never enough.

27.2 The context — historical

The historical layer answers "where are we": 200 bars of OHLCV tell the model whether price is breaking out of a range (donchian), stretched from VWAP, or at the top of a channel. It turns a raw move into a *position within a range*.

27.3 The confidence — signal intel

The intel layer answers "is this real": the recent signal history. Three agreeing signals in the window raise confidence from a blip to a pattern. This is the layer that prevents the system from trading every tick.

27.4 Worked combination

```
live:      SOXLUSDT imbalance +0.41  (buy pressure)
historical: close breaks the 55-bar high (donchian55 → buy)
intel:     three donchian55 buy signals in 90s  (pattern confirmed)
```

```
forecast = strong long → tier 1 → strategy opens
```

Each layer on its own would have been a weak or premature signal; combined they are a conviction the strategy can act on.

28. The Go code structure

What it does. Maps the code to the architecture: which module holds the bus, which holds the models, which holds the engine, and how the goroutines split the work.

Why it matters. The code should mirror the design, or the design is just a diagram. Mapping one to the other means a reader can find the model, the book, or the exit ladder in the source — and see that the decoupling is real, not aspirational.

The models and the engine run as Go binaries, split by module.

Module	Path	What it holds
`backend`	`backend/`	the market client, the flow engine, the detectors
`core`	`scalp5/core/`	the bus (stomp, topics, envelope), marketdata, recorder
`scalp`	`backend/cmd/scalp/`	the TA rules + book + stream feed (pure Go)
`scalp-bus`	`scalp5/core/cmd/scalp/`	the bus-decoupled models + engine

28.1 The `scalp-bus` binary — one process, four goroutines

```
main
 ├─ goroutine: marketdata subscription (market.candle / market.depth)
 ├─ goroutine: TA model publisher  (9 rules → model.decision)
 ├─ goroutine: depth model publisher (imbalance → model.decision)
 └─ goroutine: engine (subscribe model.decision → book → model.fill)
```

Each goroutine is independent: a slow one cannot block the others, a panic is contained, and each runs whether or not any other does. This is the runtime mirror of the bus.

28.2 The exit ladder in Go

The manage loop evaluates the four exits in order on every mark:

```
switch {
case chg >= +0.35:                    reason = "lock"
case peak >= +1.0 && chg <= peak-0.5:  reason = "trail"
case chg <= -0.5:                      reason = "stop"
case held >= 180 && chg > 0:           reason = "clock"
}
```

The order matters: profit is banked before the trail can give it back, the trail rides before the stop can cut a winner, and the stop caps any loser before the clock can let it run.

29. More worked examples

What it does. Four more scenarios: a churn day (why nothing traded), a mean-reversion trade (two models agreeing), a generation gap (a silent producer detected), and a full session replay.

Why it matters. Each example exercises a different property — the honesty of "nothing traded", the power of agreement, the detectability of loss, and the replayability of the lake. Together they show the system is not just a happy path.

29.1 A churn day — why nothing traded

Volume spikes but closes in the middle of the bar; RSI stays between 10 and 90; no range is broken. The models publish signals, but none confirm a pattern, and the book stays in `Accumulating` until the window expires.

The record shows this honestly: the WHY view lists the near-misses with their reasons, and the book's trade count is zero *for a reason* — not because the system broke, but because the market gave no confirmable pattern.

29.2 A mean-reversion trade

```
vwap_revert:  price 0.5% below VWAP → sell-side stretch → buy signal
rsi2:         RSI 8 (oversold)      → buy signal
pattern:      two agreeing buy signals in the window → confirmed
strategy:     open long, exit on lock at +0.35%
```

Two independent models agreeing on the same side is the *streak* — the confirmation that turns two weak signals into one strong tier signal.

29.3 A generation gap — detecting a silent producer

The recorder sees `model.decision` generation jump from 105 to 108. Three messages are missing. The gap is a fact, not a rounding error: a producer dropped three packets, or was silent for three cycles. The envelope's `generation` field is exactly what makes this detectable.

29.4 A full session, end to end

```
09:00  marketdata connects, depth + candles flow
09:01  models backfill 200 × 1m bars
09:02  donchian55 fires on SOLUSDT → model.decision
09:02  book records the signal in the window (1/3)
09:04  two more agree (3/3) → pattern confirmed → tier 1
09:04  strategy opens 2,500 notional
```

```
09:05  mark +0.41% → lock → close net +0.21% → model.fill
09:05  fill joins the lake, correlated to the decision
```

Every step is an envelope with an epoch timestamp, so the whole session can be replayed from the lake — which is the point of recording once and reading many times.

30. Operations and monitoring

What it does. Lists what must be running, the health checks, and the restart order.

Why it matters. A system that cannot be operated cannot be trusted. The checks (feed alive, recorder consuming, generation gaps) turn "is it working?" into a set of numeric answers, and the restart order protects the append-only lake.

30.1 What must be running

Service	Port	Purpose
market data	8796	the source of price, depth and candle envelopes
recorder	8801	consumes the bus once and writes the lake
envelope explorer	8802	the viewer for every recorded envelope
the bus	61613	STOMP broker
scalp (models + engine)	—	publishes decisions, trades

Start them in order: broker, market data, recorder, then the models and the explorer. If the explorer comes up first it will simply be empty until the recorder starts.

30.2 Health checks

Check	How	Expected
feed alive	GET :8796/health	prices > 0, kline series rising
recorder consuming	GET :8801/stats	received = written, errors 0
bus bound	queue q.rec.model.decision	consumers 1, publish rising
a model is silent	producer generation	no gap between consecutive envelopes

A generation gap is a fact: if `model.decision` jumps from 105 to 108, three packets were lost or the producer was silent for three cycles. The gap is detectable *because* the envelope carries a monotonic generation.

30.3 Restarting

Restart order matters. The explorer reads the lake the recorder writes; the models read the market data. Stop in reverse, start in order, and the record is never broken — the lake is append-only, so a restart never rewrites history.

31. Testing the models

What it does. Describes the test harness and why generated data is used.

Why it matters. A model is a pure function, so it can be tested against deterministic bars — no network, no clock, no state. A test that reads the live market is a bet, not a test. This is the discipline that makes a rule change safe to ship.

Every model is a **pure function of its inputs**, so every model is testable against generated data — no network, no clock, no state. This is what the test harness relies on.

31.1 The harness

Suite	What it verifies
`test_engine_book_models`	the book: open, mark, the exit ladder, the floor
`test_signal_scenarios`	each rule's signal against synthetic bars
`test_realistic_scenarios`	realistic price paths
`test_crazy_market`	a 192-row stress sweep — extreme moves
`table_results`	the expected-vs-actual result table

The rule: a behaviour fix ships its regression test in the same commit. A fix without a test is half-work.

31.2 Why generated data

A test that reads the live market is not a test — it is a bet on whatever the market happens to be doing. Generated data lets the same input be asserted every time: a 10× volume burst must read as a 10× burst, a –3σ drop must read as a cascade, a range breakout must read as a breakout. Deterministic inputs, deterministic assertions.

32. Appendix — the complete topic registry

What it does. Lists every topic with its plane, producers and retention.

Why it matters. This is the full producer contract surface — the complete answer to "who may write what, and how long it lives". It is the appendix you consult when adding a topic or diagnosing a missing binding.

Topic	Plane	Producers	Retention
`market.price`	market	market_data	drop
`market.candle`	market	market_data	drop
`market.depth`	market	market_data	drop
`market.mini24h`	market	market_data	drop
`ticks`	market	market_data	drop
`intel.signal`	intel	signals	60s
`intel.flow`	intel	signals	60s
`intel.movers`	intel	market_data, signals	60s
`intel.board`	intel	market_agent	60s
`intel.ta`	intel	tad	60s
`intel.forecast`	intel	forecast	60s
`alert.pumpdump`	alert	backend, signal_bus	300s
`model.decision`	model	models10, curve_rider, …	5m
`model.fill`	model	same	5m
`model.pnl`	model	same + supervisor	5m
`model.rank`	model	alpha_rank	5m
`model.evaluate`	model	tad	60s
`risk.state`	risk	risk	5m
`review.verdict`	review	portfolio_review	5m
`exec.quote`	exec	maker	60s
`system.health`	system	healthpub, master	10m
`system.control`	system	master, supervisor, cli	1d

32.1 Reading the retention column

- **drop** — live only; a subscriber must be connected to see it (no backlog, by design).
- **60s / 5m / 10m** — kept for that window so a late subscriber can catch up on state.
- **1d** — control messages persist longest, because a command must survive a slow consumer.

33. Glossary, expanded

What it does. Defines every term in one place.

Why it matters. A shared vocabulary is what lets two people read the same diagram and mean the same thing. The glossary is the contract for the words.

Term	Meaning
envelope	the fixed carrier: data, meta, references
plane	the traffic class: market, intel, model, risk, control, system
topic	the channel an envelope is published on
producer	the component that computed a payload
generation	a producer's monotonic counter; a gap means loss
correlation id	the identity that follows one piece of work
reference	a link to an input envelope, declared or derived
interceptor	a rule that runs before work and can refuse
operator	the capability that produces a payload
transformer	a stage that changes an envelope's shape
hop	one recorded pipeline stage
refusal	a decision the governor declined, with the rule named
tier signal	a strength-ranked, pattern-confirmed decision
intel window	the book's bounded rolling signal history
imbalance	bid volume minus ask volume, normalised
gross multiple	total notional divided by equity
lake	the recorded store of envelopes
streak	several agreeing signals inside the window
forecast	f(live · historical · signal intel)

34. Summary

What it does. Condenses the whole document to one paragraph and three principles.

Why it matters. The summary is the answer to "what is this, in one breath": a decoupled streaming forecast, where books hold the intel, the bus keeps it real-time, and the window keeps it informed.

The system is a **decoupled streaming forecast**: market data streams in (never polled), the models publish numbers (never trading directly), the bus carries them as stamped envelopes (never backlogged), the books accumulate the intel (never too much), and the strategies trade only on confirmed patterns (never on a lone blip).

- **A model alone is an opinion; a book with history is a conviction.**
- **Data alone is noise; data + context + memory is a forecast.**
- **The bus keeps it decoupled and real-time; TTL keeps it fresh; the window keeps it informed.**

That is the architecture, and every diagram in this document is one view of the same principle: **combined, historical, and continuously updated — so it can forecast, and fire a tier signal worth trading.**

Appendix D — the payloads, verbatim

A `market.depth` packet:

```
{"symbol":"BTCUSDT","bids":[{"p":81777.5,"q":2.13}],
 "asks":[{"p":81778.0,"q":19.9}],
 "bid_qty":2.13,"ask_qty":19.9,"imbalance":-0.81,"ts":1792741261000}
```

A `model.decision` packet:

```
{"symbol":"SOLUSDT","side":"buy","model":"donchian55",
 "score":1.0,"ts_ms":1792741261000,"why":"close >= 55-bar high"}
```

A `model.fill` packet:

```
{"model":"donchian55","symbol":"SOLUSDT","side":"buy",
 "reason":"lock","net_pct":0.21,"pnl_usd":5.25,"ts_ms":1792741267000}
```

Each is a small, self-describing number — the exact kind of packet the bus was designed to carry, and the exact kind of evidence the lake preserves.

Appendix E — the planes, their purpose and their producers

Plane	Purpose	Example producers
`market`	raw exchange data, live only	market_data
`intel`	derived intelligence	signals, tad, forecast
`model`	decisions and outcomes	models10, curve_rider, alpha_rank
`risk`	the governor's state and refusals	risk
`review`	portfolio verdicts	portfolio_review
`exec`	execution quotes	maker
`control`	orchestration and commands	supervisor, master
`system`	health and control	healthpub, master

The plane is the first thing any reader checks: it tells you *what class of traffic* a packet is before you read its payload — market data, a model's opinion, or the governor's decision. Together with the topic, the plane is the address; the envelope is the letter.

Appendix F — THE X FILES

> "The truth is out there." In this system, the truth is **in the envelopes**. Every decision, every trade, every refusal is a stamped packet in an append-only lake — so any question an investigator can ask, the record can answer. This appendix is the field manual: how to **debug a decision**, how to **optimise a model**, and how the **absolute traceability** of the platform works. It is the most satisfying part of the whole design.

F.1 The investigation mindset

When a model traded — or did not — the first question is never "is it a bug". It is:

1. **What did it read?** — the references, with their ages at execution.
2. **What did it decide?** — the verdict, the confidence, the reason.
3. **What stopped it?** — the refusal, with the rule and the number.

The system records all three for every decision, so the investigation is a *read*, not a *guess*. There is no "it probably did X" — there is only the envelope that says what it did.

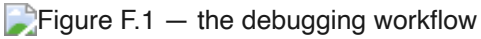


Figure F.1 — every decision resolves to a verdict you can reproduce from the lake.

F.2 Debugging a decision

F.2.1 Traded or refused?

The first fork: did the decision become a trade, or a refusal? The answer decides which view to open.

- **Refused** → the WHY view, which ranks refusals by rule and shows the exact reason.
- **Traded** → the EXECUTION view, which shows what the model read and how old each input was.

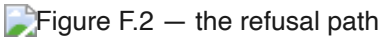


Figure F.2 — a refusal names the rule and the number, and is itself an envelope in the lake.

F.2.2 What the model actually read

The execution trace answers the narrowest, most useful question: *for this one fill, which raw envelopes did the model read, in what order, and how old were they when it acted?*

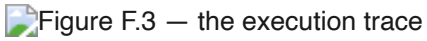


Figure F.3 — inputs are listed with their age at execution; a stale input is a finding, not a feeling.

Finding	Meaning
input 0.2s old	fresh — the decision was well-founded
input 3.2s old	still fresh for most topics
input 3200s old	stale — the model acted on old data

F.2.3 A worked refusal

```
model.decision NEARUSDT  side=buy  confidence=0.62
  verdict    REFUSED
  reason     stale signals: 3230s old (max 90s)
  pipeline   authority pass · keying pass · correlation pass · freshness pass
  refs       market.candle 0.5s · market.depth 0.2s (both fresh)
```

The freshness interceptor passed — the *envelope* was fresh. The refusal came from the model's own signal-age check. The market data was seconds old; the *signals* were 53 minutes old. The investigation lands on the real fault: a signal producer was silent for that symbol, and the model correctly refused to act on stale information.

F.3 Optimising the models

Optimisation is a loop, not a one-off. The same decoupling that makes the system observable makes it tuneable: because every model is a pure function and every outcome is a stamped packet, a parameter change can be measured against the exact same inputs.

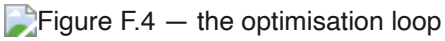


Figure F.4 — evaluate, tune, re-run on generated data, re-evaluate.

F.3.1 The measure

Metric	What it tells you
realised PnL	the bottom line
win rate	how often the model is right
fee drag	how much the 0.2% fee eats the edge
mean hold	whether exits fire as intended
largest loss	whether the stop is actually bounding losses

F.3.2 The probation and rotation

A model earns its place by surviving probation: its first N minutes of live trading. If the PnL falls below the floor, it is stopped, its post-mortem is written, and another model is tried.

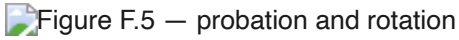


Figure F.5 — survive probation or be stopped; a stopped model is post-mortemed and replaced.

F.3.3 Tuning the forecast

The forecast is a function of three layers; tuning is calibrating their weights. The tier signal is the feedback: a weight change that raises the quality of tier-1 signals is kept, one that does not is reverted.

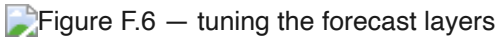


Figure F.6 — the tier signal feeds back into the layer weights.

F.3.4 A worked tuning

`rsi2` was buying at RSI < 10 and losing to fees. The investigation showed the winners were real but the losers were fee-sized. Tuning the threshold from <10 to <5 cut the low-quality entries, and the win rate rose. The change was measured against the *same* generated bars before and after — so the improvement is a number, not an opinion.

F.4 Absolute traceability

This is the fascinating part: **the platform is evidence**. Nothing is lost, nothing is inferred silently, and every step is linked by one correlation id.

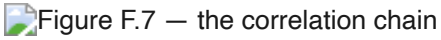


Figure F.7 — one cid follows the work from market tick to fill to pnl, into an append-only lake.

F.4.1 Why this is absolute

Property	What it means
append-only lake	history is never rewritten
monotonic generation	a gap is detectable, never hidden
correlation ids	every step links to its parent
derived refs are labelled	inferred lineage is never passed off as declared
refusals are recorded	"why not" is evidence, not a gap
case hash	two reports are comparable by fingerprint

F.4.2 The case file

For any decision, the system can produce a **case file**: the verdict, the inputs with ages, the pipeline stages, and a case hash that fingerprints the scope — so two copies can be compared and a changed one detected.

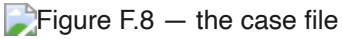


Figure F.8 — a case file is the decision, its inputs, its pipeline, and a hash.

F.4.3 Reconstructing a trade from a correlation id

```
$ resolve cid-8c50fc9388443f4e

  market.candle  SOLUSDT  09:02:01.123   (the trigger)
  model.decision SOLUSDT  buy  cid-8c50fc9388443f4e
  model.fill     SOLUSDT  buy  net +0.21%  parent cid-8c50fc9388443f4e
  model.pnl      donchian55  equity 5005.25  cid-8c50fc9388443f4e
```

From one id, the entire life of a trade is reassembled — the tick that triggered it, the decision, the fill, and the resulting pnl. That is what "absolute traceability" means: the record is complete enough to replay the trade, and honest enough to show its own gaps.

F.5 The X FILES, closed

The truth is out there — and in this system it is *literally* out there, in the lake, as a chain of stamped envelopes. Debugging is reading that chain. Optimising is measuring it. Traceability is the fact that the chain is complete and append-only. That is the reward for building decoupled: **every question is answerable, and every answer is evidence.**

35. The journey of market data

What it does. Traces market data from the exchange WebSocket all the way to a recorded envelope and a model's decision. It is the *artery* of the system: everything downstream consumes this one stream.

Why it matters. If you understand this one journey you understand the platform's shape: one source of market data, one bus, one recorder, many consumers. Everything else is a consumer of this same stream.

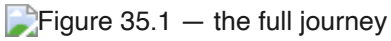


Figure 35.1 — Binance → marketdata → bus → recorder → lake, and bus → models → book → fills.

35.1 The marketdata pipeline



Figure 35.2 — WebSocket frames are filtered at the source into three signals, then published.

The marketdata service does three things to every frame, in order: it **filters** (keeps only the fields each consumer needs — the BPF idea), it **collapses** (the depth becomes one imbalance number, the candles a bounded cache), and it **publishes** (an envelope per signal onto the bus). Nothing is stored in the service itself — it is a pure pump.

35.2 The depth journey

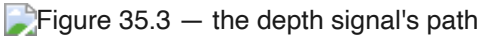


Figure 35.3 — the order book becomes one imbalance number, published and consumed.

35.3 The candle journey

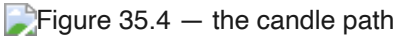


Figure 35.4 — candles are seeded once (history), then streamed, and the TA models read the cache.

The candle history is **backfilled once** — 200 bars per symbol — because the bus carries only newly-closed candles (TTL, no backlog). The models read the seeded cache plus the live updates, so a model always has the *context* (history) and the *now* (the latest bar).

35.4 The marketdata subscription

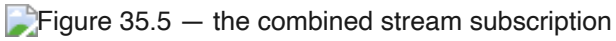


Figure 35.5 — 90 symbols over one combined stream, dispatched by stream name.

35.5 Ticks to signal

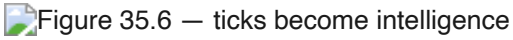


Figure 35.6 — raw ticks are enriched into intel.signal, which the models consume as context.

36. The bus in detail

What it does. Moves every packet between every component, with an exchange per plane and a queue per topic, and fans one publisher out to many subscribers.

Why it matters. The bus is the *single integration point*. Understanding its shape — exchanges, queues, bindings, fan-out — is understanding how any component talks to any other without a single direct call.

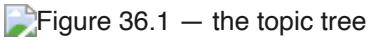


Figure 36.1 — exchanges group the planes; queues deliver to consumers.

36.1 Fan-out: one publisher, many subscribers



Figure 36.2 — one marketdata publish reaches the recorder, the models and the book at once.

This is the decoupling made concrete: the publisher does not know the recorder, the models or the book exist. It publishes once; the broker fans the packet out to every subscriber. Adding a new consumer is subscribing, not wiring.

36.2 A binding

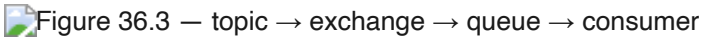


Figure 36.3 — the topic binds to an exchange, which feeds a queue, which delivers to a consumer.

36.3 The envelope lifecycle

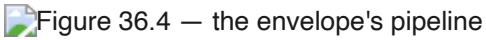


Figure 36.4 — producer → interceptors → operator → transformer → lake, with refusals recorded.

36.4 The recorder fan-out



Figure 36.5 — the recorder writes the journal and the index, and fans out to the explorer and dashboards.

36.5 The bus, as a sequence

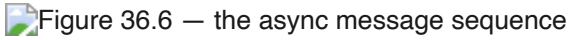


Figure 36.6 — one tick's message sequence, end to end, with no component blocking another.

37. Models subscribe, asynchronously

What it does. A model is a subscriber: it receives the latest market state, computes a number, and publishes a decision — all without blocking, and all without knowing who listens.

Why it matters. Asynchrony is the runtime twin of decoupling. A model that blocks would slow the pipeline; a model that panics would kill it. Because each model subscribes, computes and publishes independently, neither can happen.

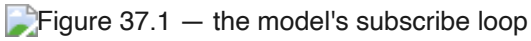


Figure 37.1 — subscribe, receive the latest state, compute, publish, repeat.

37.1 The subscription lifecycle

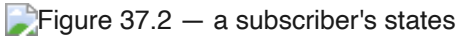


Figure 37.2 — disconnected → connecting → subscribed → receiving, with reconnect on error.

37.2 The async flow across goroutines



Figure 37.3 — each goroutine is a producer, a consumer, or both; none blocks the others.

37.3 A decision, asynchronously

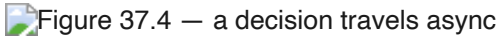


Figure 37.4 — tick → model → decision → book → strategy → fill, each hop asynchronous.

37.4 The book subscribes and confirms

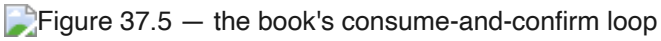


Figure 37.5 — the book records the decision in its window and trades only on a confirmed pattern.

37.5 The supervisor, async



Figure 37.6 — the supervisor launches and watches, but never makes a trading decision.

38. Correlation and the explorer

What it does. One correlation id follows every piece of work; the explorer reads the lake and renders the whole journey — the flow, the trace, the why.

Why it matters. This is where the architecture pays off for a human: the explorer turns the append-only lake into a view of *what happened and why*, resolved by correlation id.

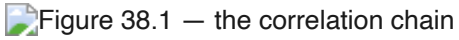


Figure 38.1 — one cid links the trigger, the decision, the fill and the pnl.

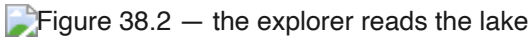


Figure 38.2 — the explorer is a read-only viewer: ENVELOPES, JOURNEY, EXECUTION, WHY, GANTT, BOOKS, HEALTH.

15. Field dictionary and glossary

15.1 Envelope metadata

Field	Type	Meaning
`version`	int	envelope contract revision
`plane`	string	market, intel, model, risk, control, system
`topic`	string	the bus topic
`key`	string	the symbol, when it concerns one
`producer`	string	the component that computed it
`generation`	uint	the producer's counter; gaps mean loss
`cid`	string	correlation id
`computed_at`	timestamp	when produced — staleness gates use this
`received_at`	timestamp	when the recorder accepted it
`refs`	array	inputs with topic, key, generation, age_ms
`hops`	array	pipeline stages with outcome

15.2 Payload fields by topic

Topic	Fields
`model.decision`	engine, symbol, side, allowed, reason, confidence
`model.fill`	engine, symbol, side, entry, exit, gross_pct, net_pct, peak_pct, reason, held_s, notional, pnl_usd
`market.depth`	symbol, bids[], asks[], bid_qty, ask_qty, imbalance
`market.candle`	symbol, interval, t, o, h, l, c, v, q, Q
`risk.state`	snapshot{equity, gross_multiple, longs, shorts, per_symbol}

15.3 Glossary

Term	Meaning
envelope	the fixed carrier: data, meta, references
interceptor	a rule that runs before work, and can refuse
operator	the capability that produces a payload
transformer	a stage that changes an envelope's shape
correlation id	the identity that follows one piece of work
tier signal	a strength-ranked, pattern-confirmed decision
intel window	the book's bounded rolling signal history
lake	the recorded store of envelopes
imbalance	bid volume minus ask volume, normalised